MIL-F-11206G(MU)
6 November 1970
SUPERSEDING
MIL-F-11206F(MU)
25 September 1968

MILITARY SPECIFICATION

FUZE, BOMB, AN-M173A1

1. SCOPE

1.1 This specification covers one type of fuze.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

STANDARDS

FEDERAL

FED. STD. NO. 595 - Colors.

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-331 - Fuze and Fuze Components, Environmental and Performance Test for.
MIL-STD-810 - Environmental Test Methods.
MIL-STD-1168 - Lot Numbering of Ammunition.

DRAWINGS

US ARMY MUNITIONS COMMAND

EDGEWOOD ARSENAL

DL14-15-636 - Fuze, Bomb, AN-M173A1.
IDL14-15-636 - Fuze, Bomb, AN-M173A1, Inspection Data List.

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PUBLICATIONS

EDGEWOOD ARSENAL INSPECTION STANDARD

R7-4 - Radiographic Evaluation of Fuze, Bomb, AN-M173A1.
R14-13 - Soundness and Surface Condition of Zinc Die Cast Components for Fuze AN-M173A1 and Fuze FMU-7C/B.

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials and components.

3.1.1 Materials. All materials cited on Drawing DL14-15-636 or on the subsidiary drawings shall conform to the specifications listed htereon or to the specific characteristics set forth on the drawings.

3.1.2 Components. All components of the fuze shall conform to the specifications and drawings listed on Drawing DL14-15-636 and subsidiary drawings.

3.2 Assembly. The fuze shall be assembled as shown on Drawing C14-15-636 (see 6.3).

3.3 Safety. The fuze less the detonator and booster cup shall not function when dropped 5 feet ± 1 inch, primer end down when tested as specified in 4.4.4.1.

3.4 Striker assembly (Drawing C14-15-909). The striker assembly shall comply with the following when tested as specified in 4.4.4.2.

(a) A 0.087 inch diameter steel pin with a hemispherical end shall pass between the balls freely.

(b) A 0.095 inch diameter steel pin with a hemispherical end shall not pass between the balls.

(c) The balls shall move freely by their own weight when the striker assembly is rotated about the longitudinal axis when that axis is held in the horizontal position.

3.5 Vane arming assembly (Drawing B14-15-276). The arming pin shall show no movement in the vane when a force of 250 ± 10 pounds is applied for 15 ± 1 second when tested as specified in 4.4.4.3.

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3.6 *Radiographic examination*. The fuze, less the detonator and booster cup, shall be examined radiographically as specified in 4.4.4.4.

3.7 Arming. The fuze less the booster cup shall arm in a maximum period of 2 seconds when subjected to a wind velocity of 110 miles per hour. Armed fuzes which will be subjected to the functioning test shall meet the requirements of 3.12 prior to being tested as specified in 4.4.4.5 (see 6.5).

3.8 Functioning. The armed fuze having met the requirements of 3.12 and 3.7 shall function upon impact when dropped 6 feet + 2 inches with random orientation when tested as specified in 4.4.4.6 (see 6.3 and 6.5).

3.9 Jolt. The assembled fuze, without detonator and tetryl pellets, shall comply with the requirements of MIL-STD-331 when tested as specified in 4.4.4.7.

3.10 Jumble. The assembled fuze, without detonator and tetryl pellets, packaged in accordance with 5.1, shall comply with the requirements of MIL-STD-331 when tested as specified in 4.4.4.8.

3.11 Forty-foot drop. The assembled fuze, without detonator and tetryl pellets, when dropped at random orientation, shall comply with the requirements of MIL-STD-331 when tested as specified in 4.4.4.9.

3.12 Vibration. The fuze shall comply with 3.7 and 3.8 subsequent to vibration when tested as specified in 4.4.4.10.

3.13 Torque. There shall be no movement between the head and the case and between the case and the booster cup when a breakaway torque of 40 inch-pounds maximum is applied and tested as specified in 4.4.4.11 (see 6.4).

3.14 Soundness and surface condition. The case and booster cup shall be examined as specified in 4.4.4.12.

3.15 Preproduction. Prior to the start of regular production, a preproduction lot of fuzes, without detonators and tetryl pellets, shall be produced in accordance with this specification for examination and tests (see 4.3).

3.16 Workmanship. The fuze shall be free from foreign matter (oil, grease or dirt) and damage such as dents, cracks or deformation which could affect its intended use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.

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4.1.1 Supplier's responsibility. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to specified requirements.

4.1.2 Objective evidence. The supplier shall provide objective evidence acceptable to the contracting officer that the requirements of 3.1 and section 5 for which specific inspection has not been provided in this specification have been satisfied.

4.2 Classification of inspections.

(a) Preproduction inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

4.3 Preproduction inspection.

4.3.1 Lot. A preproduction lot of 250 fuzes shall be manufactured using the same methods, materials, equipment and processes as will be used during regular production.

4.3.2 Sampling.

4.3.2.1 For examination and nondestructive tests.

(a) Each fuze shall be taken for critical defects.
(b) Each striker assembly shall be taken for test.
(c) Fifty fuzes shall be taken at random for the classification of defects.
(d) Each fuze shall be taken for arming test, however the fifty fuzes used in 4.3.3.2 shall be vibrated prior to arming.

4.3.2.2 For destructive tests. Eighty fuzes shall be taken at random from the preproduction lot.

4.3.3 Inspection procedure.

4.3.3.1 For examination and nondestructive tests.

(a) Each fuze shall be tested in accordance with 4.4.4.1, 4.4.4.4 and 4.4.4.5(a).
(b) Each striker assembly shall be tested in accordance with 4.4.4.2.
(c) The sample fuzes shall be examined and tested in accordance with the classification of defects (4.4.3.4). The AQL's do not apply.

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(d) Each fuze shall be tested in accordance with 4.4.4.5(b), however the fifty fuzes shall be tested in accordance with 4.4.4.10 prior to 4.4.4.5(b). Accepted fuzes may be resafed and used for other tests.

4.3.3.2 For destructive tests. Sample fuzes shall be tested in accordance with table I.

Table I. Preproduction destructive testing

Quantity	Requirement	Test
*50	3.12	4.4.4.10
	3.7	4.4.4.5
	3.8	4.4.4.6
10	3.9	4.4.4.7
10	3.10	4.4.4.8
10	3.11	4.4.4.9

*Test fuzes in accordance with 4.4.4.10, then test in accordance with 4.4.4.5 and 4.4.4.6.

4.3.4 Acceptance/rejection criteria. The preproduction lot and sample shall comply with the examinations and tests specified in 4.3.3 to be acceptable. The acceptance number is zero. The supplier shall obtain written approval from the contracting officer prior to proceeding with regular production.

4.4 Quality conformance inspection.

4.4.1 Lotting. Each lot shall be identified and controlled in accordance with MIL-STD-1168. Contained in any one interfix number of fuzes, no more than one interfix number from one manufacturer of the following components is permitted.

(a) Primers
(b) Detonators
(c) Tetryl pellets

4.4.2 Sampling.

4.4.2.1 For examination and nondestructive tests. Sampling shall be conducted in accordance with MIL-STD-105. Sampling for vibration shall be in accordance with MIL-STD-105, inspection level S-4.

4.4.2.2 For destructive tests. For functioning test, the same sample fuzes taken for vibration test, shall then be armed and functioned. Sampling for jolt, jumble and 40 foot drop, shall be conducted in accordance with MIL-STD-105, inspection level S-2.

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4.4.3 Inspection procedure.

4.4.3.1 For examination and nondestructive tests. The sample fuzes shall be examined and tested in accordance with the classification of defects and MIL-STD-105. The sample fuzes shall be vibrated in accordance with 4.4.4.10 and armed in accordance with 4.4.4.5.

4.4.3.2 For destructive tests.

(a) Functioning. The sample fuzes shall be tested in accordance with 4.4.4.6 and MIL-STD-105, using an AQL of 1.0 percent defective.

(b) Jolt, jumble and 40 foot drop. The sample fuzes shall be tested in accordance with 4.4.4.7, 4.4.4.8, 4.4.4.9 and MIL-STD-105, using an AQL of 1.5 percent defective.

4.4.3.3 For critical defects. Each item in the lot shall be inspected for the critical characteristics listed in the classification of defects.

4.4.3.4 Classification of defects. (See inspection data list IDL14-15-636 for inspection equipment required to perform the examinations specified in this section.)

(a) Fuze, bomb, AN-M173A1 (Drawing C14-15-636).

Categories	Defects	Acceptance standards
Critical:		
1	Fuze safety	4.4.4.1
2	Safety pin formed incorrect	4.4.4.5(a)
3	Radiographic examination	4.4.4.4
Major:	AQL 0.65 percent defective	
101	Torque incorrect	4.4.4.11
102	Thread (1-1/2-24NS-2) incorrect	
103	Arming time incorrect	
104	Workmanship (3.16)	
	AQL 1.0 percent defective	
105	Marking missing or incorrect	

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4.4.3.4 Classification of defects (continued).

(b) Sleeve assembly (Drawing C14-15-907).

Categories		Defects
Critical:		None defined
Major:	AQL 0.65 percent defective	
101		Primer inverted
102		Surface of primer deformed

(c) Striker assembly (Drawing C14-15-909).

Categories		Defects	Acceptance standards
Critical:		None defined	
Major:	AQL 0.65 percent defective		
101		Striker point profile incorrect	
102		Striker assembly incorrect	4.4.4.2
103		Length incorrect	

(d) Spring, striker (Drawing B14-15-40).

Categories		Defects
Critical:		None defined
Major:	AQL 0.65 percent defective	
101		Load at 0.156 inch height incorrect

(e) Vane, arming assembly (Drawing B14-15-276).

Categories		Defects	Acceptance standards
Critical:		None defined	
Major:	AQL 0.65 percent defective		
101		Arming pin pushout	4.4.4.3
102		Arming pin protrusion incorrect	

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4.4.3.4 Classification of defects (continued).

(f) Case (Drawing C14-15-277) and cup, booster (Drawing C14-15-278).

Categories	Defects	Acceptance standards
Critical:	None defined	
Major:	AQL 0.65 percent defective	
101	Surface and soundness condition incorrect	4.4.4.12

(g) Pin, arming (Drawing B14-15-194).

Categories	Defects
Critical:	None defined
Major:	AQL 0.65 percent defective
101	Arming pin diameter incorrect
102	Arming pin length incorrect

4.4.4 Tests. Tests shall be conducted as follows: (See inspection data list IDL14-15-636 for inspection equipment required to perform the tests specified in this section.)

4.4.4.1 Safety. Prior to assembly of the detonator and tetryl pellets, the unarmed fuze shall be tested as follows:

(a) Turn the arming vane down until it shoulders on the fuze body.

(b) Drop the fuze, primer end down, through a tube of suitable diameter onto a rubber pad 1/4 $\pm$ $^{1}/_{0}^{8}$ inch thick having a durometer hardness of 40 $\pm$ 10 and supported on the entire lower surface by a 1 inch thick steel plate.

4.4.4.2 Striker assembly. The assembly shall be checked for adequate staking by inserting first a 0.087 inch diameter pin gage and second a 0.095 inch diameter pin gage into the 0.097 inch diameter hole. After gaging, the assembly shall be rotated and movement of the balls observed.

4.4.4.3 Vane arming assembly. The pin in the arming vane assembly shall withstand the specified force exerted vertically downward on the centerline of the pin.

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4.4.4.4 Radiographic examination. The fuze less detonator and booster cup shall be examined and evaluated in accordance with Edgewood Arsenal Inspection Standard R7-4.

4.4.4.5 Arming.

(a) The safety pin shall be extracted.

(b) The fuze shall be placed in a wind tunnel or test device in such a manner that the flat face of one of the blades will be normal to the air stream. The test section shall have a minimum diameter of 4.5 inches and shall have laminar airflow throughout. The time required for the arming vane to turn 40 revolutions shall be recorded. Forty revolutions of the arming vane equals 1/2 inch lateral movement of the vane.

4.4.4.6 Functioning. The fuze shall be dropped with random orientation to impact upon a steel plate or concrete slab.

4.4.4.7 Jolt. The fuze shall be tested in accordance with test 101 of MIL-STD-331.

4.4.4.8 Jumble. The fuze shall be tested in accordance with test 102 of MIL-STD-331.

4.4.4.9 Forty-foot drop. The fuze shall be tested in accordance with test 103 of MIL-STD-331, free fall, except the fuze shall be dropped without a bomb.

4.4.4.10 Vibration. The fuze shall be tested in accordance with method 514, table 514-I, equipment category g (shipment by common carrier), procedure X, curve AB of MIL-STD-810.

4.4.4.11 Torque. The fuze shall be placed in a fixture and the specified load applied against the fuze case and booster cup.

4.4.4.12 Soundness and surface condition. The case and booster cup shall be examined and evaluated in accordance with Edgewood Arsenal Inspection Standard R14-13.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Level shall be as specified (see 6.2).

5.1.1 Level A. The fuze shall be packaged one per can in accordance with Drawing D4-2-81.

5.1.2 Level C. The fuze shall be packaged in accordance with 5.1.1.

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5.2 Packing. Level shall be as specified (see 6.2).

5.2.1 Level A. The fuze, packaged in accordance with 5.1.1, shall be packed 50 per box in accordance with Drawing D4-2-81.

5.2.2 Level C. The fuze, packaged in accordance with 5.1.1, shall be packed in accordance with 5.2.1, except the box is not required to be preservative treated, filler pads may be domestic class and strapping may be class A or C.

5.3 Marking. In conjunction with information provided in the contract or order and information developed by the contractor, marking shall be as specified herein.

5.3.1 Unit containers. Each unit container shall be marked with information consisting of the Federal stock number, item identification (nomenclature), quantity and unit of issue, level of packaging and date packaged, and the lot number. Marking shall be in accordance with MIL-STD-129. In addition, each container shall be provided with a 1/2 inch wide band of color yellow number 23538 or 33538 conforming to FED STD No. 595. The band shall be located in a position that will not interfere with other prescribed markings.

5.3.2 Exterior containers. Each box shall be marked in accordance with Drawing D14-15-995.

5.3.3 Palletized load (see 5.4). When items are palletized, the unit load shall be marked in accordance with Drawing D14-15-994.

5.4 Palletization. When specified in the contract or order (see 6.2) the packed fuzes shall be palletized in accordance with Drawing D14-15-994.

6. NOTES

6.1 Intended use. This item is intended for use in initiating a fire bomb.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) The Government will supply radiographic standards.
(c) Preproduction.

(1) Time allowed for suppliers submission of samples for Government test and evaluation after award of contract.
(2) Name and address of test facility and shipping instructions when testing is performed by the Government.
(3) Time required for the Government to notify the supplier whether to proceed with production.

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(d) When items are required to be palletized (mandatory for level A).

(e) When preservative treatment is required for wood components of palletized load (mandatory for level A).

(f) Appropriate address marking.

(g) Appropriate identification marking.

(h) Level of packaging and packing required.

6.3 Caution. Suppliers are advised that every precaution should be taken to protect personnel and prevent explosions while preparing mixtures and assembling and testing fuzes. Skin contact with tetryl should be avoided.

6.4 Torque. Prior to test, the sealant should be cured.

6.5 Safety. For safety the detonator may be installed into the fuze after the arming test. As a further safety the arming pin may be reinserted into the fuze prior to installing the detonator and shall be removed prior to functioning.

Custodian:

Army - MU

Preparing activity:

Army - MU(EA)

Project No. 1325-A395

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426 REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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Commanding Officer
Edgewood Arsenal
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Edgewood Arsenal, Maryland 21010

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